

GWENNAP HEAD, United Kingdom

WMO: 038060

Lat: 50.0369N

Lon: 5.6805W

Elev: 63

StdP: 100.57

Time Zone: 0.00 (EUW)

Period: 82-92

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	0.1	1.4	-3.6	2.8	2.0	-2.4	3.1	2.8	26.4	5.6	23.4	6.3	8.8	60	0.835

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	3.2	21.0	16.8	19.8	16.6	18.8	16.1	17.9	19.5	17.2	18.8	16.7	18.2	5.1	60

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.2	12.4	18.3	16.6	11.9	18.0	16.1	11.5	17.5	50.5	19.5	48.6	19.0	47.2	18.3	21.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 20.4	18.4	15.8	DB	-1.6	23.4	2.5	2.4	-3.3	25.1	-4.8	26.5	-6.2	27.9	-8.0	29.6	(4)
(5)			WB	-2.8	19.0	2.2	1.3	-4.4	19.9	-5.7	20.6	-6.9	21.4	-8.5	22.3	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	11.0	7.7	6.6	7.7	8.6	11.1	13.3	15.8	15.9	14.4	12.4	9.6	8.8	(6)
(7)		DBStd	3.74	2.52	2.91	1.66	1.90	2.05	1.75	1.94	1.79	1.61	1.57	2.31	2.06	(7)
(8)		HDD10.0	381	74	96	71	50	9	0	0	0	0	2	34	45	(8)
(9)		HDD18.3	2673	330	328	328	293	225	151	83	79	117	183	262	294	(9)
(10)		CDD10.0	755	2	1	1	7	42	99	179	182	133	77	22	9	(10)
(11)		CDD18.3	7	0	0	0	0	0	1	4	2	0	0	0	0	(11)
(12)		CDH23.3	7	0	0	0	0	0	0	6	1	0	0	0	0	(12)
(13)		CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0	(13)
(14)	Wind	WSAvg	8.0	10.0	9.8	8.7	7.7	7.2	6.4	6.4	6.1	6.8	8.6	9.2	9.7	(14)
(15)	Precipitation	PrecAvg	1002	117	90	63	67	58	60	68	76	69	100	124	111	(15)
(16)		PrecMax	1249	197	187	141	156	140	137	214	190	170	167	248	235	(16)
(17)		PrecMin	808	26	16	20	13	7	12	7	18	8	35	43	32	(17)
(18)		PrecStd	134	48	48	31	38	31	37	43	44	38	36	48	54	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	11.7	11.7	11.6	16.1	20.1	21.4	24.1	22.3	20.2	16.3	13.7	13.1	(19)
(20)			MCWB	11.0	10.4	9.9	11.5	14.9	15.8	18.3	17.6	17.0	14.8	12.9	12.1	(20)
(21)		2%	DB	11.3	11.1	10.7	13.8	17.7	19.0	21.6	20.4	18.6	15.5	13.2	12.3	(21)
(22)			MCWB	10.5	10.2	9.6	10.5	13.7	14.9	17.2	17.0	16.3	14.2	12.4	11.6	(22)
(23)		5%	DB	10.9	10.6	10.2	12.3	15.7	17.3	20.2	19.3	17.6	14.9	12.9	11.9	(23)
(24)			MCWB	10.1	9.8	9.3	10.1	12.2	14.3	16.9	16.8	15.9	13.7	12.1	11.2	(24)
(25)		10%	DB	10.6	10.1	9.9	11.1	14.0	16.0	18.9	18.5	16.8	14.5	12.5	11.4	(25)
(26)			MCWB	9.7	9.4	9.0	9.6	11.3	13.8	16.1	16.4	15.4	13.3	11.7	10.6	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.6	11.1	10.6	12.2	15.4	16.6	19.5	18.5	17.7	15.7	13.3	12.4	(27)
(28)			MCDWB	11.6	11.2	11.0	14.6	19.0	20.0	21.8	21.0	18.7	16.0	13.4	12.7	(28)
(29)		2%	WB	11.0	10.5	10.1	11.2	13.9	15.4	18.0	17.9	17.0	14.7	12.9	12.0	(29)
(30)			MCDWB	11.1	10.8	10.4	12.6	17.2	17.6	20.3	19.2	18.0	15.0	13.1	12.2	(30)
(31)		5%	WB	10.5	10.1	9.7	10.6	12.7	14.8	17.2	17.4	16.4	14.2	12.5	11.5	(31)
(32)			MCDWB	10.8	10.4	10.0	11.6	15.1	16.5	19.4	18.6	17.2	14.6	12.7	11.7	(32)
(33)		10%	WB	10.1	9.6	9.3	10.1	11.9	14.2	16.6	16.9	15.8	13.7	12.0	11.0	(33)
(34)			MCDWB	10.4	10.0	9.7	10.9	13.3	15.5	18.5	18.1	16.5	14.2	12.4	11.3	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	2.9	2.9	2.8	3.8	3.7	3.4	3.7	3.2	3.3	2.6	2.9	2.6	(35)
(36)			MCDBR	2.7	2.5	2.9	5.5	6.4	6.1	5.9	4.6	4.2	2.8	2.3	2.3	(36)
(37)			MCWBR	3.4	2.9	2.8	3.9	4.0	4.1	4.0	3.0	3.3	2.7	2.5	2.9	(37)
(38)		5% WB	MCDBR	2.7	2.3	2.5	4.1	5.2	4.9	5.0	3.7	3.4	2.4	2.2	2.4	(38)
(39)			MCWBR	3.5	3.0	2.8	3.3	3.6	3.6	3.7	2.8	3.1	2.7	2.7	3.1	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.339	0.352	0.379	0.383	0.391	0.377	0.383	0.367	0.364	0.360	0.337	0.332	(40)	
(41)		taud	2.357	2.382	2.331	2.357	2.381	2.451	2.441	2.501	2.491	2.453	2.403	2.320	(41)	
(42)		Ebn at Noon	686	780	821	861	868	882	869	869	833	763	685	629	(42)	
(43)		Edh at Noon	68	84	104	112	114	108	107	96	88	77	64	61	(43)	
(44)	All-Sky Solar Radiation	RadAvg	0.85	1.64	2.79	4.54	5.47	5.99	5.50	4.73	3.59	2.07	1.09	0.68	(44)	
(45)		RadStd	0.07	0.15	0.30	0.30	0.40	0.55	0.37	0.32	0.24	0.13	0.08	0.07	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (19 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page